Data Quality Assessment & Rule Catalogue

Analytical findings from profiling every file in **__initial_load** (2026-11-13). Section 5 is the reusable rule catalogue: every rule has an ID that matches **data_quality_validation.sql**, and a **baseline** (failing count found during initial-load profiling) so later runs can be compared against it.

1. Dataset inventory

Table (target)	Source files	Rows	Cols	Grain / primary key	Source system
region_master	country-master/region_master.csv	5	7	region_code	MDM_CORE
country_master	country-master/country_master.csv	35	20	country_code	MDM_CORE
tax_master	country-master/tax_master.csv	35	9	tax_code	TAX_ENGINE
currency_master	country-master/currency_master.csv	27	9	currency_code	TREASURY_SYS
product_category_master	product-master/product_category_master.csv	10	8	category_code	PLM_CORE
product_family_master	product-master/product_family_master.csv	43	8	family_code	PLM_CORE
product_model_master	product-master/product_model_master.csv	111	9	model_code	PLM_CORE
product_sku_master	product-master/product_sku_master.csv	650	8	sku_code	ERP_SKU
product_country_availability	product-master/product_country_availability.csv	22,750	8	(sku_code, country_code)	MDM_PRODUCT
store_master	store-master/store_master.csv	121	22	store_code	RETAIL_OPS
customer_master	customer-master/<yyyy>/<cc>/*.csv (245 files)	260,513 (104,469 customers)	26 (+snapshot _year)	(customer_id, snapshot_year)	MDM
sales_header	sales-transaction/<yyyy>/sales_header_<yyyy>.csv (7)	642,546	15	transaction_sk (alt: transaction_id)	SAP_SD
sales_item	sales-transaction/<yyyy>/sales_item_<yyyy>.csv (7)	642,546	11	transaction_line_id	-

2. Column-level profile highlights

Table	Column	Profile
country_master	population_millions / gdp_usd_billions	5-1,428 (avg 130) / 249-27,360 (avg 2,542); no nulls
country_master	market_tier / retail_store_supported	Tier1 12, Tier2 23 / Y 24, N 11
tax_master	tax_type / tax_rate	VAT 25, GST 5, SALES_TAX, IVA, ICMS, SST, NONE / 0.0-0.255 (avg 0.155); inclusive Y 33, N 2 (US, HK)
currency_master	minor_unit	2 for 25 currencies, 0 for JPY/KRW; 17 distinct symbols for 27 currencies
product_family_master	launch_year	2019-2024; IPH/MAC/WCH/AIR/ACC 6 families each, IPD 5, HMP 3, TV 2, DSP 2, VPR 1
product_model_master	launch_date / discontinue_date	2019-02-09 to 2024-11-08 / 100% null
product_sku_master	price_tier / variant	Premium 306, Ultra 255, Standard 89 / 112 distinct, 2 formats
product_country_availability	local_launch_date / is_available	2019-02-09 to 2024-12-06, 0-32 day lag / 100% 'Y'
store_master	store_open_date / floor_area_sqft / annual_rent_usd	2016-04-29 to 2026-04-10 / 5,205-24,570 / 0.66M-19.4M
store_master	state_code	Null for 39 stores in 16 countries without state data

Table	Column	Profile
customer_master	date_of_birth / registration_date	1950-04-18 to 2008-04-16 / 2019-01-01 to 2025-12-31
customer_master	customer_segment / loyalty_tier	Consumer 75%, Business 15%, Education 10% / None 50%, Silver 23%, Gold 16%, Platinum 12%
customer_master	is_active	'True' 94%, 'False' 6% (string)
sales_header	channel_id / payment_method	POS 67%, ONLINE 33% / 9 methods ~11.1% each
sales_header	gross_amount / total_discount / net_total	29.01-4,996.32 (median 393) / 70% zero, max 488.89 / 27.96-6,123.52 (median 436)
sales_header	transaction_timestamp	2019-01-01 00:49 to 2026-01-01 03:07; Q4 peak; flat weekdays
sales_item	quantity / category_code	1 (79%), 2 (21%) / IPH 186K, ACC 145K, AIR 128K, TV 58K, IPD 48K, HMP 29K, WCH 27K, MAC 20K, VPR 1.3K, DSP 1.2K

3. Relationship (referential integrity) results

Child	Parent	Cardinality	Orphans	Note
country_master.region_code	region_master	N:1	0	All regions used
country_master.currency_code	currency_master	N:1	0	EUR shared by 9
country_master.tax_code	tax_master	1:1	0	Prefix = country_code
product_family_master.category_code	product_category_master	N:1	0	
product_model_master.family_code	product_family_master	N:1	0	1-6 models per family
product_sku_master.model_code	product_model_master	N:1	0	1-18 SKUs per model
product_country_availability.sku_code / country_code	sku / country_master	N:1	0	Full cross join (GAP-011)
store_master.country_code / region_code	country / region_master	N:1	0	24 countries
store_master.tax_jurisdiction_code	tax_master	N:1	121 stores (70 codes)	GAP-021
customer_master.country_code	country_master	N:1	0	All 35 countries
sales_header.customer_id	customer_master	N:1	0	85,567 buyers
sales_header.store_id	store_master	N:1	0	POS only
sales_header.country_code / currency	country / currency_master	N:1	0	Currency = country currency
sales_item.transaction_sk	sales_header	1:1	0	Single-line baskets
sales_item.sku_code	product_sku_master	N:1	0	All 650 SKUs sold
sales_item (sku, header.country)	product_country_availability	N:1	0	

4. Key analytical findings

• Amounts are USD-scale but labelled with 27 local currencies: the same iPhone 11 SKU sells for ~700-1,600 in USD, JPY, KRW and INR alike; 66,607 JPY/KRW sales have decimals although minor_unit = 0.

• All arithmetic reconciles exactly (0 exceptions): line totals, header-to-line sums and net = gross - discount + tax.

• Tax is added on top in 33 countries flagged tax-inclusive (451,922 sales); 5 countries use rates different from tax_master (CA 13%, BR 12%, CH 7.7%, MY 6%, FI 24%; 45,671 sales); no rate history anywhere.

• Timeline violations: 152,808 POS sales before store opening (83 stores); 111,385 sales before customer registration; 6,737 sales before SKU launch.

• customer_master is a yearly snapshot (2.5 rows per customer); grain must include snapshot_year to avoid inflated counts.

• Reference and product lifecycles are static (no discontinued products, no closed stores, no tax/region history); availability is a full cross join.

• Patterns indicating synthetic data: uniform payment mix, ONLINE Cash payments, flat weekday volume, 82,824 POS sales between 00:00-05:59, unstable unit prices (7,419 of 15,707 SKU/currency pairs vary > 2x), fake addresses and coordinates.

5. Data quality rule catalogue (baseline = initial load)

Status: PASS if failed_count = 0 (or > 0 where the expectation is '> 0'); otherwise FAIL / GAP. The scorecard query at the end of the SQL file produces this table automatically and reports change versus baseline.

Rule ID	Table	Dimension	Severity	Rule	Baseline	Expected	Status
DQ-001	region_master	Uniqueness	High	region_code is unique	0	= 0	PASS
DQ-002	country_master	Uniqueness	High	country_code is unique	0	= 0	PASS
DQ-003	tax_master	Uniqueness	High	tax_code is unique	0	= 0	PASS
DQ-004	currency_master	Uniqueness	High	currency_code is unique	0	= 0	PASS
DQ-005	country_master	Integrity	High	region_code exists in region_master	0	= 0	PASS
DQ-006	country_master	Integrity	High	currency_code exists in currency_master	0	= 0	PASS
DQ-007	country_master	Integrity	High	tax_code exists in tax_master	0	= 0	PASS
DQ-008	country_master	Uniqueness	Medium	iso_alpha3 and country_name are unique	0	= 0	PASS
DQ-009	country_master	Validity	Medium	country_code is ISO 3166-1 alpha-2 ('UK' should be 'GB')	1	= 0	FAIL
DQ-010	reference masters	Validity	High	effective_start_date <= effective_end_date	0	= 0	PASS
DQ-011	tax_master	Validity	High	tax_rate between 0 and 1 (stored as fraction)	0	= 0	PASS
DQ-012	tax_master	Timeliness	High	Tax validity covers the earliest sale date	35	= 0	FAIL
DQ-013	currency_master	Validity	Medium	minor_unit in (0,2,3)	0	= 0	PASS
DQ-014	country_master	Validity	Medium	Y/N flags contain only 'Y' or 'N'	0	= 0	PASS
DQ-015	country_master	Consistency	Low	apple_fiscal_segment is 1:1 with region_code	0	= 0	PASS
DQ-016	country_master	Consistency	Medium	AE/SA mapped to EMEA (business review)	2	= 0 after review	FAIL
DQ-020	product masters	Uniqueness	High	PKs unique: category_code, family_code, model_code, sku_code	0	= 0	PASS
DQ-021	product_country_availability	Uniqueness	High	(sku_code, country_code) is unique	0	= 0	PASS
DQ-022	product_family_master	Integrity	High	category_code exists in product_category_master	0	= 0	PASS
DQ-023	product_model_master	Integrity	High	family_code exists in product_family_master	0	= 0	PASS
DQ-024	product_sku_master	Integrity	High	model_code exists in product_model_master	0	= 0	PASS
DQ-025	product_country_availability	Integrity	High	sku_code exists in product_sku_master	0	= 0	PASS
DQ-026	product_country_availability	Integrity	High	country_code exists in country_master	0	= 0	PASS
DQ-027	product_family_master	Consistency	Medium	family_code = category_code || launch_year	0	= 0	PASS
DQ-028	product_model_master	Consistency	Medium	model_code starts with family_code	0	= 0	PASS
DQ-029	product_model_master	Consistency	Medium	YEAR(launch_date) = family launch_year	0	= 0	PASS
DQ-030	product_sku_master	Consistency	Medium	global_launch_date = model launch_date	0	= 0	PASS
DQ-031	product_country_availability	Consistency	Medium	local_launch_date >= SKU global_launch_date	0	= 0	PASS
DQ-032	model / availability	Validity	Medium	discontinue date not before launch date	0	= 0	PASS

Rule ID	Table	Dimension	Severity	Rule	Baseline	Expected	Status
DQ-033	product_model_master	Uniqueness	Medium	model_name is unique	1	= 0	FAIL
DQ-034	product_model_master	Completeness	High	Lifecycle populated: models with a discontinue_date (gap if 0)	0	> 0 expected (review)	GAP
DQ-035	product_country_availability	Consistency	High	Missing SKU x country pairs (0 = full cross join, gap)	0	> 0 expected (review)	GAP
DQ-036	product_country_availability	Uniqueness	Medium	local_part_number unique within a country	0	= 0	PASS
DQ-037	product_sku_master	Validity	Low	price_tier in (Standard, Premium, Ultra)	0	= 0	PASS
DQ-038	product_sku_master	Validity	Low	sku_code matches M + 6 alnum + LL/A	0	= 0	PASS
DQ-040	store_master	Uniqueness	High	store_code is unique	0	= 0	PASS
DQ-041	store_master	Integrity	High	country_code exists in country_master	0	= 0	PASS
DQ-042	store_master	Integrity	High	region_code exists in region_master	0	= 0	PASS
DQ-043	store_master	Consistency	Medium	store region_code = country region_code	0	= 0	PASS
DQ-044	store_master	Consistency	Low	store_code prefix = country_code	0	= 0	PASS
DQ-045	store_master	Integrity	High	tax_jurisdiction_code exists in tax_master	121	= 0	FAIL
DQ-046	store_master	Consistency	Medium	No stores in countries with retail_store_supported = 'N'	0	= 0	PASS
DQ-047	store_master	Validity	Medium	store_close_date >= store_open_date	0	= 0	PASS
DQ-048	sales_header	Timeliness	High	POS transactions on/after store_open_date	152,808	= 0	FAIL
DQ-049	store_master	Timeliness	Medium	SCD effective_start_date <= store_open_date	121	= 0	FAIL
DQ-050	store_master	Validity	Medium	latitude in [-90,90], longitude in [-180,180]	0	= 0	PASS
DQ-051	store_master	Validity	Medium	Plausible rent: annual_rent_usd / floor_area_sqft <= 1,000	29	= 0	FAIL
DQ-052	store_master	Validity	Low	format_code in (FLG, MALL, MINI)	0	= 0	PASS
DQ-053	store_master	Validity	Low	address_line1 without embedded newline	5	= 0	FAIL
DQ-054	store_master	Consistency	Low	US tax_jurisdiction_code = 'US_' || state_code || '_STD'	0	= 0	PASS
DQ-060	customer_master	Uniqueness	High	(customer_id, snapshot_year) is unique	0	= 0	PASS
DQ-061	customer_master	Consistency	High	customer_id <-> customer_number is 1:1	0	= 0	PASS
DQ-062	customer_master	Integrity	High	country_code exists in country_master	0	= 0	PASS
DQ-063	customer_master	Consistency	Low	region / country_name / preferred_language match country_master	0	= 0	PASS
DQ-064	customer_master	Validity	Medium	customer_number format CUST-<acq_year>-<CC>-<6 digits>, consistent with acquisition_year and country_code	0	= 0	PASS
DQ-065	customer_master	Consistency	Medium	acquisition_year = YEAR(registration_date)	0	= 0	PASS
DQ-066	customer_master	Consistency	Low	full_name = first_name || ' ' || last_name	0	= 0	PASS
DQ-067	customer_master	Validity	Medium	email has a valid format	0	= 0	PASS
DQ-068	customer_master	Uniqueness	Medium	email not shared by different customers (count of shared emails)	6,067	= 0	FAIL

Rule ID	Table	Dimension	Severity	Rule	Baseline	Expected	Status
DQ-069	customer_master	Validity	Medium	Age at registration >= 13 (rows)	3,652	= 0	FAIL
DQ-070	customer_master	Validity	Low	street_address has no leading/trailing whitespace	2,850	= 0	FAIL
DQ-071	customer_master	Validity	Low	city has no leading/trailing whitespace	16	= 0	FAIL
DQ-072	customer_master	Consistency	Medium	customer_type = NEW exactly in the acquisition snapshot	0	= 0	PASS
DQ-073	customer_master	Completeness	Medium	No gaps in yearly snapshot presence (customers)	0	= 0	PASS
DQ-074	customer_master	Consistency	Medium	Immutable attributes unchanged across snapshots (name, DOB, email, segment, country)	0	= 0	PASS
DQ-075	customer_master	Consistency	Low	No loyalty_tier downgrades between consecutive snapshots	0	= 0 (info)	PASS
DQ-076	customer_master	Validity	Low	Domain values: gender, customer_segment, loyalty_tier, customer_type	0	= 0	PASS
DQ-080	sales_header	Uniqueness	High	transaction_sk and transaction_id are unique	0	= 0	PASS
DQ-081	sales_item	Uniqueness	High	transaction_line_id and (transaction_sk, line_number) are unique	0	= 0	PASS
DQ-082	sales_item	Integrity	High	Every item has a header	0	= 0	PASS
DQ-083	sales_header	Completeness	High	Every header has at least one item	0	= 0	PASS
DQ-084	sales_item	Completeness	Low	Transactions with more than one line (0 = single-line baskets, gap)	0	> 0 expected (info)	GAP
DQ-085	sales_header	Integrity	High	customer_id exists in customer_master	0	= 0	PASS
DQ-086	sales_header	Integrity	High	store_id exists in store_master	0	= 0	PASS
DQ-087	sales_header	Integrity	High	country_code exists in country_master	0	= 0	PASS
DQ-088	sales_header	Integrity	High	currency exists in currency_master	0	= 0	PASS
DQ-089	sales_item	Integrity	High	sku_code exists in product_sku_master	0	= 0	PASS
DQ-090	sales_header	Consistency	High	POS requires store_id; ONLINE must have no store_id	0	= 0	PASS
DQ-091	sales_header	Consistency	High	POS country_code = store country_code	0	= 0	PASS
DQ-092	sales_header	Consistency	High	currency = country currency_code	0	= 0	PASS
DQ-093	sales_header	Consistency	Medium	Sale country = customer country	0	= 0	PASS
DQ-094	sales_item	Consistency	Medium	category_code matches SKU hierarchy	0	= 0	PASS
DQ-095	sales_item	Accuracy	High	line_total = quantity*unit_price - discount_amount + tax_amount (+/-0.01)	0	= 0	PASS
DQ-096	sales_header	Accuracy	High	Header gross/discount/tax/net = sum of item lines (+/-0.01)	0	= 0	PASS
DQ-097	sales_header	Accuracy	High	net_total = gross_amount - total_discount + total_tax (+/-0.01)	0	= 0	PASS
DQ-098	sales_header / sales_item	Validity	High	No negative/zero amounts; discount <= gross; quantity > 0	0	= 0	PASS
DQ-099	sales_header	Accuracy	High	Effective tax rate = tax_master rate for the country (+/-0.002)	45,671	= 0	FAIL
DQ-100	sales_item	Timeliness	High	Sale on/after SKU global_launch_date	6,737	= 0	FAIL
DQ-101	sales_item	Timeliness	High	Sale on/after local_launch_date in that country	6,739	= 0	FAIL
DQ-102	sales_item	Integrity	High	SKU is available in the sale country	0	= 0	PASS
DQ-103	sales_header	Timeliness	High	Sale on/after customer registration_date	111,385	= 0	FAIL

Rule ID	Table	Dimension	Severity	Rule	Baseline	Expected	Status
DQ-104	sales_header	Consistency	Medium	No sales by customers inactive in that year's snapshot	40,311	= 0 (review)	FAIL
DQ-105	sales_header	Consistency	Low	Customer present in the snapshot of the sale year	54	= 0	FAIL
DQ-106	sales_header	Validity	High	No fractional amounts in zero-decimal currencies (JPY, KRW)	66,607	= 0	FAIL
DQ-107	sales_header	Consistency	High	Tax-inclusive countries must not add tax on top of gross	451,922	= 0 (decision needed)	FAIL
DQ-108	sales_header	Validity	Medium	ONLINE orders not paid in Cash	23,488	= 0	FAIL
DQ-109	sales_item	Consistency	Medium	SKU+currency price spread (max/min) <= 2x (count of pairs)	7,419	= 0 (review)	FAIL
DQ-110	sales_header	Validity	Low	POS transactions outside 06:00-23:59 store hours	82,824	= 0 (review)	FAIL
DQ-111	sales_header	Timeliness	Medium	No future-dated transactions	0	= 0	PASS
DQ-112	sales_item	Validity	Low	quantity in expected range (1-10)	0	= 0	PASS

6. Baseline scorecard

100 rules: **73 PASS**, **24 FAIL**, **3 GAP** (expected to be non-zero but zero).

Dimension	Rules	Not passing
Accuracy	4	1
Completeness	4	2
Consistency	28	6
Integrity	19	1
Timeliness	7	6
Uniqueness	14	2
Validity	24	9

7. Load recommendations for validation to work

• File format: TYPE=CSV, SKIP_HEADER=1, FIELD_OPTIONALLY_ENCLOSED_BY="" (store addresses contain newlines), ENCODING='UTF8', EMPTY_FIELD_AS_NULL=TRUE, TRIM_SPACE=FALSE (so whitespace rules can detect issues; trim in curated layer).

• customer_master: add snapshot_year from METADATA$FILENAME (e.g. TO_NUMBER(SPLIT_PART(METADATA$FILENAME,'/',-3))).

• sales_header/sales_item: union the year files; partition/cluster by transaction_timestamp::DATE, not by file year.

• Types: NUMBER(12,2) for money, NUMBER(5,4) for tax_rate, DATE for dates, TIMESTAMP_NTZ(6) for created_at/updated_at, VARCHAR for postal codes and phone numbers.

• Schedule the scorecard query after each load (or attach equivalent Data Metric Functions) and alert when change_vs_baseline > 0.